

Structural Causal Models

an introduction

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DAG-recap

In past lectures on DAGs

1. causal directed acyclic graphs (DAGs) encode assumptions on what variables cause what
2. an intervention is defined as a mutilation of this DAG where the treatment variable no longer 'listens' to its parents
3. a causal effect is the effect of an intervention
4. DAG patterns:
 - fork (confounding)
 - chain (mediation)
 - collider
5. typically: condition on confounders, don't condition on mediators or colliders
6. in more complex DAGs, use d-separation to check identifyability
7. backdoor criterion

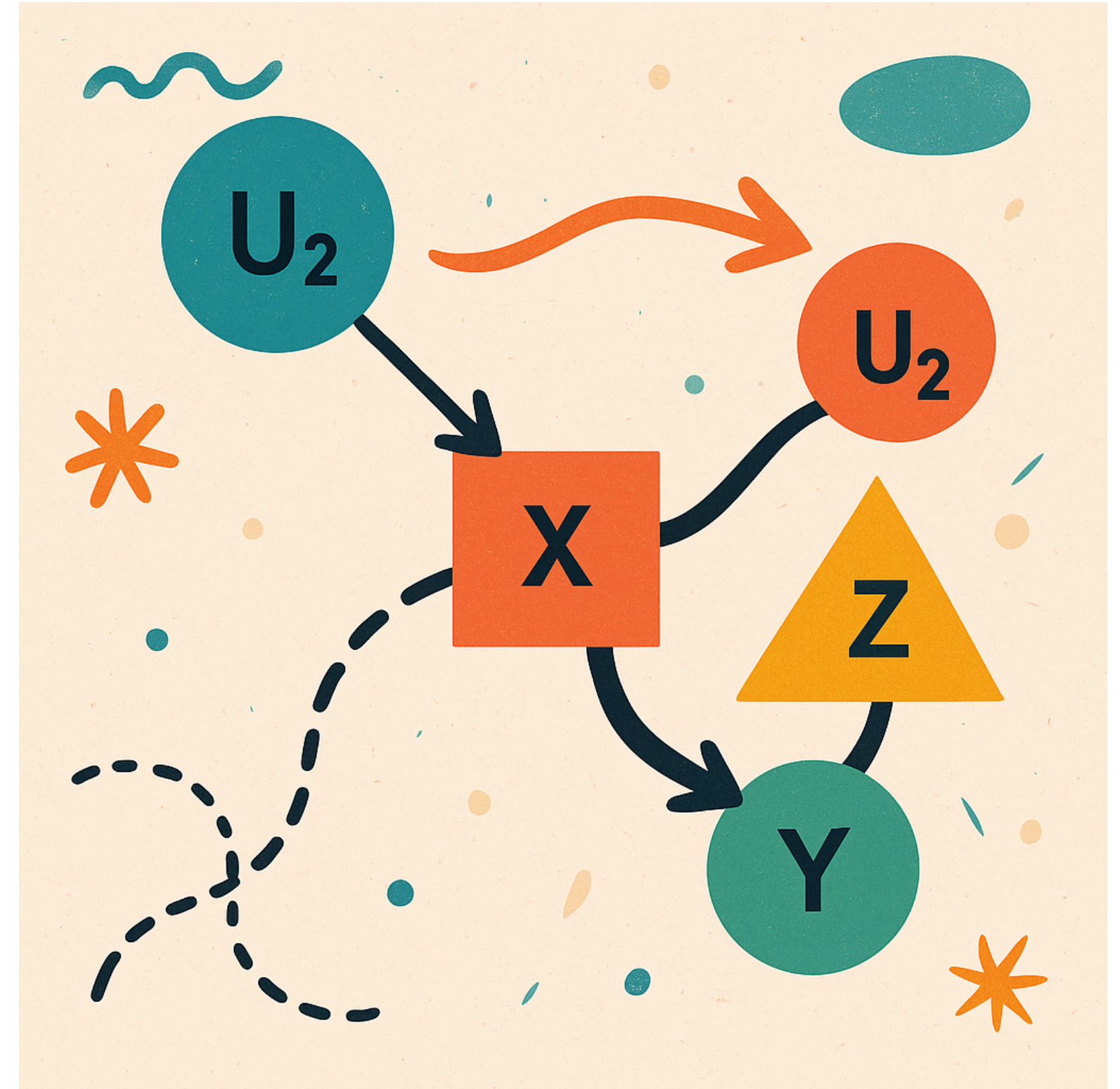
In this lecture: Structural Causal Models (SCMs)

$$U_Z, U_T, U_Y \sim p(U)$$

$$Z = f_Z(U_Z)$$

$$T = f_T(Z, U_T)$$

$$Y = f_Y(T, Z, U_Y)$$





Why SCMs?

- With DAGs we can:
 - express (non-parametric) prior knowledge
 - understand that seeing $\neq$ doing
 - know what variables to condition on for estimating treatment effect
- However,
 - DAGs and RCTs do not cover all causal questions
 - SCMs go a level deeper than DAGs, DAGs naturally ‘arise’ from SCMs
 - some questions are not identified when only specifying a DAG, but we may have additional information that can lead to identification
 - understand ‘identifyability’
 - SCM thinking aligns with physical thinking about the world and is a natural way to think about causality



Topics of today

- SCMs: the world as computer programs
- interventions are submodels
- bonus queries:
 - counterfactuals
- Pearl Causal Hierarchy
- reflections on DAGs, limitations



Structural Causal Models: definitions

Think of the world as a computer program with a set of

- (endogenous) *variables*:
 - `surgery` = duration of surgery (hours)
 - `los` = length of stay in hospital post surgery (days)
 - `survival` = survival time (years)
- *background variables* (exogenous, `unobserved`):
 - `u_surgery`, `u_los`, `u_survival`
- *functions* `f_` for each *variable* which depend on its *parents* `pa_` and its own *background* `u_`:
 - `surgery = f_surgery(pa_surgery, u_surgery)`
 - `los = f_los(pa_los, u_los)`
 - `survival = f_survival(pa_survival, u_survival)`

Together these define a *Structural Causal Model* (see definition 7.1.1 in [Pearl 2009](#), and further)
(notation: $M = \langle U, V, F \rangle$)

Structural Causal Model 1: an R program

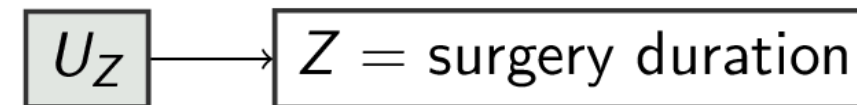
```
1 f_surgery <- function(u_surgery) { # pa_surgery = {}
2   u_surgery
3 }
4 f_los <- function(surgery, u_los) { # pa_los = {surgery}
5   surgery + u_los
6 }
7 f_survival <- function(surgery, los, u_survival) { # pa_survival = {surgery, los}
8   survival = los - 2 * surgery + u_survival
9 }
10
11 scm1 <- function(u_surgery, u_los, u_survival) {
12   surgery = f_surgery(u_surgery)
13   los = f_los(surgery, u_los)
14   survival = f_survival(surgery, los, u_survival)
15   c(surgery=surgery, los=los, survival=survival)
16 }
17 scm1(2, 1, 5)
```

surgery	los	survival
2	3	4

Recursive Structural Causal Models imply a Directed Acyclic Graph

An SCM is *recursive*, i.e. *acyclic* when following the chain of parents, you never end up at the same variable twice

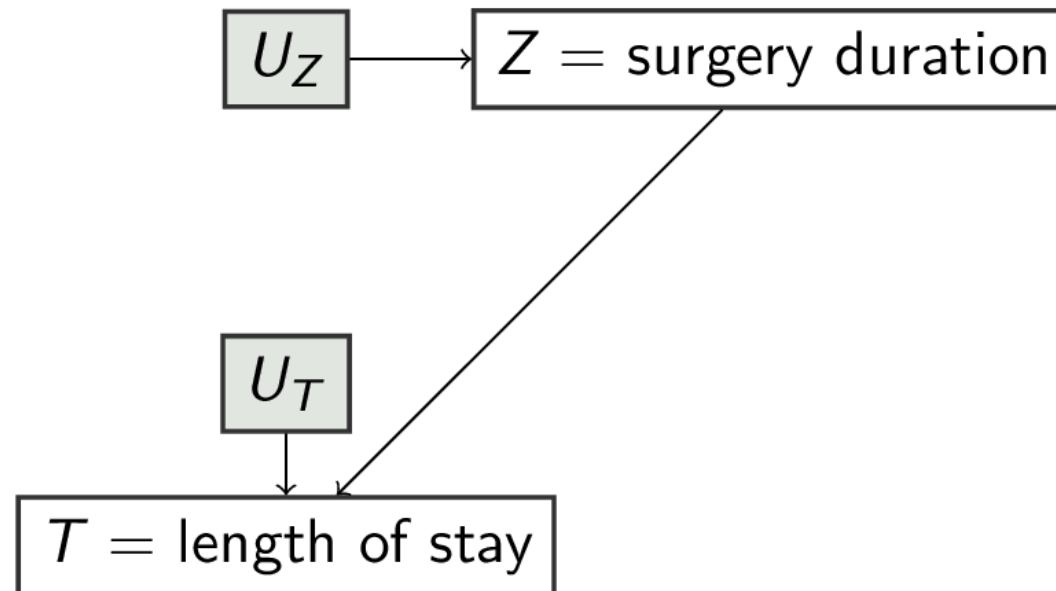
```
1 scm1 <- function(u_surgery, u_los, u_survival) {  
2   surgery = f_surgery(u_surgery)  
3   los     = f_los(surgery, u_los)  
4   survival = f_survival(surgery, los, u_survival)  
5   c(surgery=surgery, los=los, survival=survival)  
6 }
```



Recursive Structural Causal Models imply a Directed Acyclic Graph

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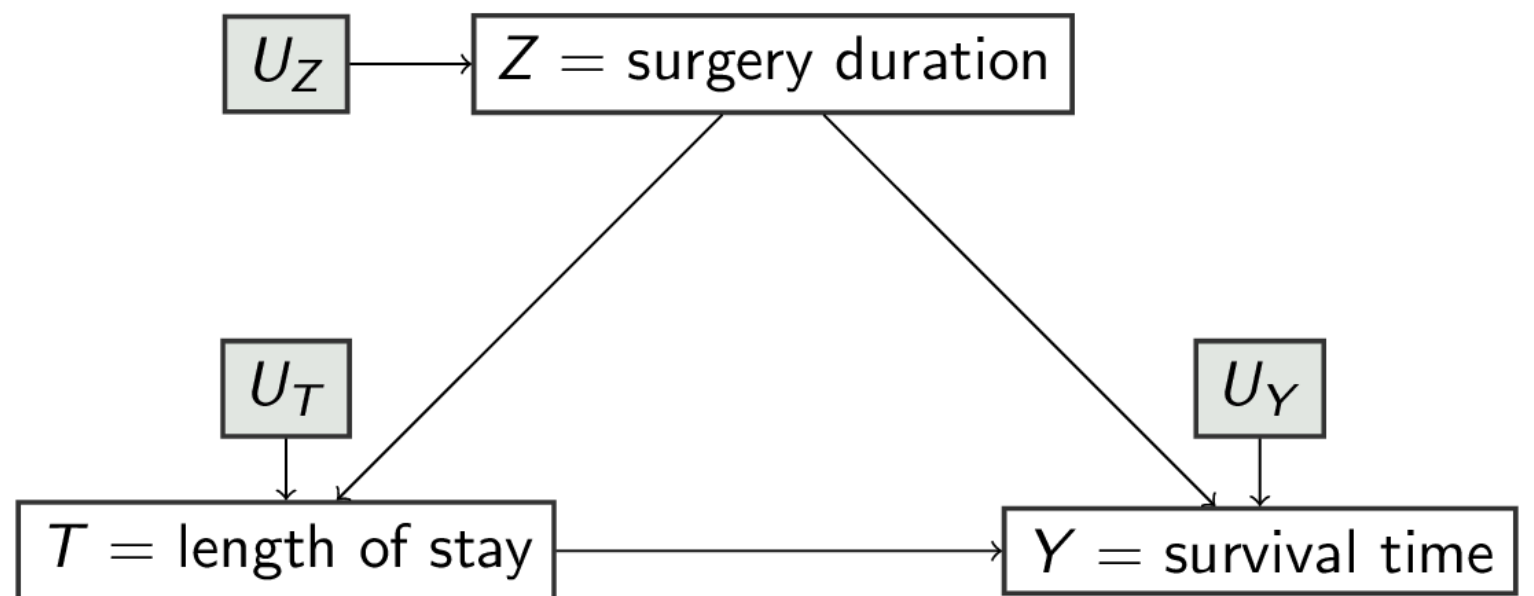
```
1 scm1 <- function(u_surgery, u_los, u_survival) {  
2   surgery = f_surgery(u_surgery)  
3   los     = f_los(surgery, u_los)  
4   survival = f_survival(surgery, los, u_survival)  
5   c(surgery=surgery, los=los, survival=survival)  
6 }
```



Recursive Structural Causal Models imply a Directed Acyclic Graph

An SCM is *recursive*, i.e. *acyclic* when following the chain of parents, you never end up at the same variable twice

```
1 scm1 <- function(u_surgery, u_los, u_survival) {  
2   surgery = f_surgery(u_surgery)  
3   los     = f_los(surgery, u_los)  
4   survival = f_survival(surgery, los, u_survival)  
5   c(surgery=surgery, los=los, survival=survival)  
6 }
```



`scm1` (without specifying the `f_s`) and the DAG are equivalent (they describe the same knowledge of the world)

for the remainder, we assume recursiveness

Submodel and Effect of Action

- **submodel:** in `scm1` replace `f_los` with a specific *value*, e.g. `7` days

```
1 submodel7 <- function(u_surgery, u_los, u_survival) {  
2   surgery = f_surgery(u_surgery)  
3   los = 7  
4   survival = f_survival(surgery, los, u_survival)  
5   c(surgery=surgery, los=los, survival=survival)  
6 }  
7  
8 submodel7(2, 1, 5)
```

surgery	los	survival
2	7	8

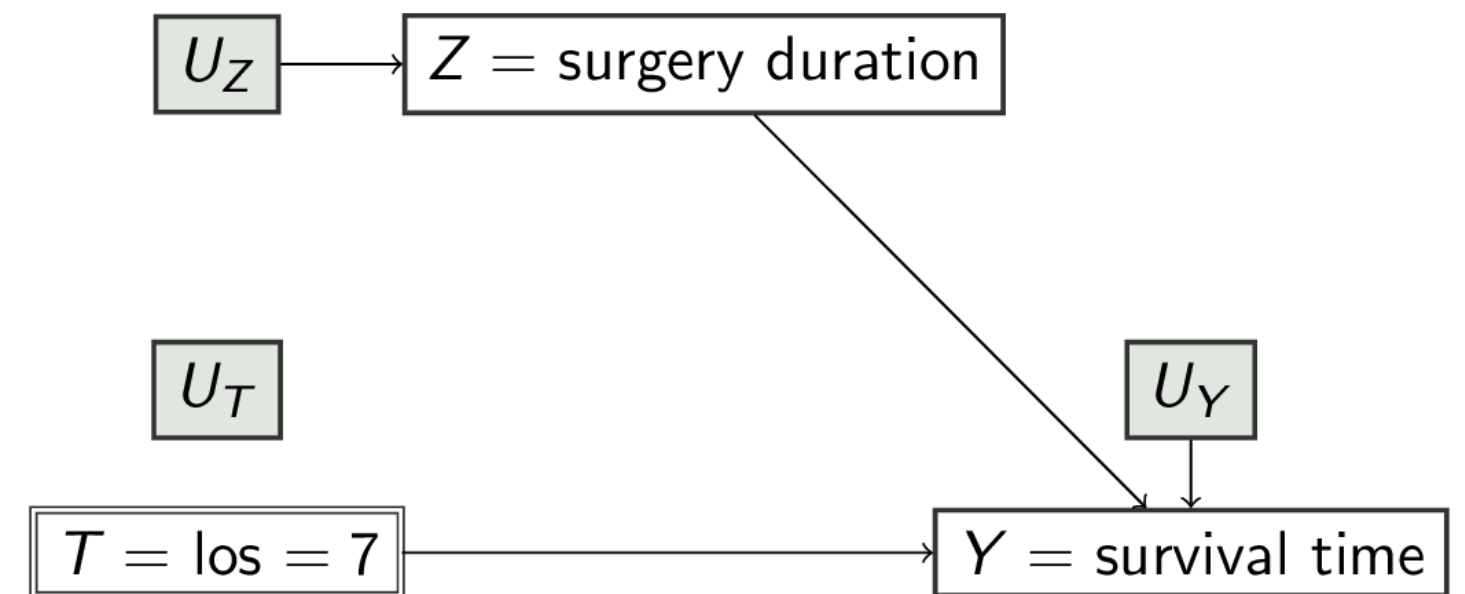
- **effect of action:** resulting SCM of submodel (notation: $M_x = \langle U, V, F_x \rangle$)

Submodel and Effect of Action as a mutilated DAG

In `scm1` replace `f_los` with a specific *value*, e.g. 7 days (notation: M_x)

```
1 submodel7 <- function(u_surgery, u_los, u_survival) {  
2   surgery = f_surgery(u_surgery)  
3   los = 7  
4   survival = f_survival(surgery, los, u_survival)  
5   c(surgery=surgery, los=los, survival=survival)  
6 }  
7  
8 submodel7(2, 1, 5)
```

surgery	los	survival
2	7	8



The DAG describes a submodel where T no longer ‘listens’ to any variables but is controlled to be equal to a specific value (e.g. 7)

The *Effect of Action* $do(X = x)$ is defined as the submodel M_x .

Specifying a distribution for exogenous variables U

- Exogenous variables **U** represent random variation in the world.
- We can specify a *distribution* for them (e.g. Gaussian, Uniform)

```
1 sample_u <- function() {  
2   u_surgery = runif(1, 2, 8)  
3   u_los      = runif(1, -1, 7)  
4   u_survival = runif(1, 8, 13)  
5   c(u_surgery=u_surgery, u_los=u_los, u_survival=u_survival)  
6 }  
7 sample_u()
```

```
u_surgery    u_los u_survival  
2.066870    3.846598 8.949930
```

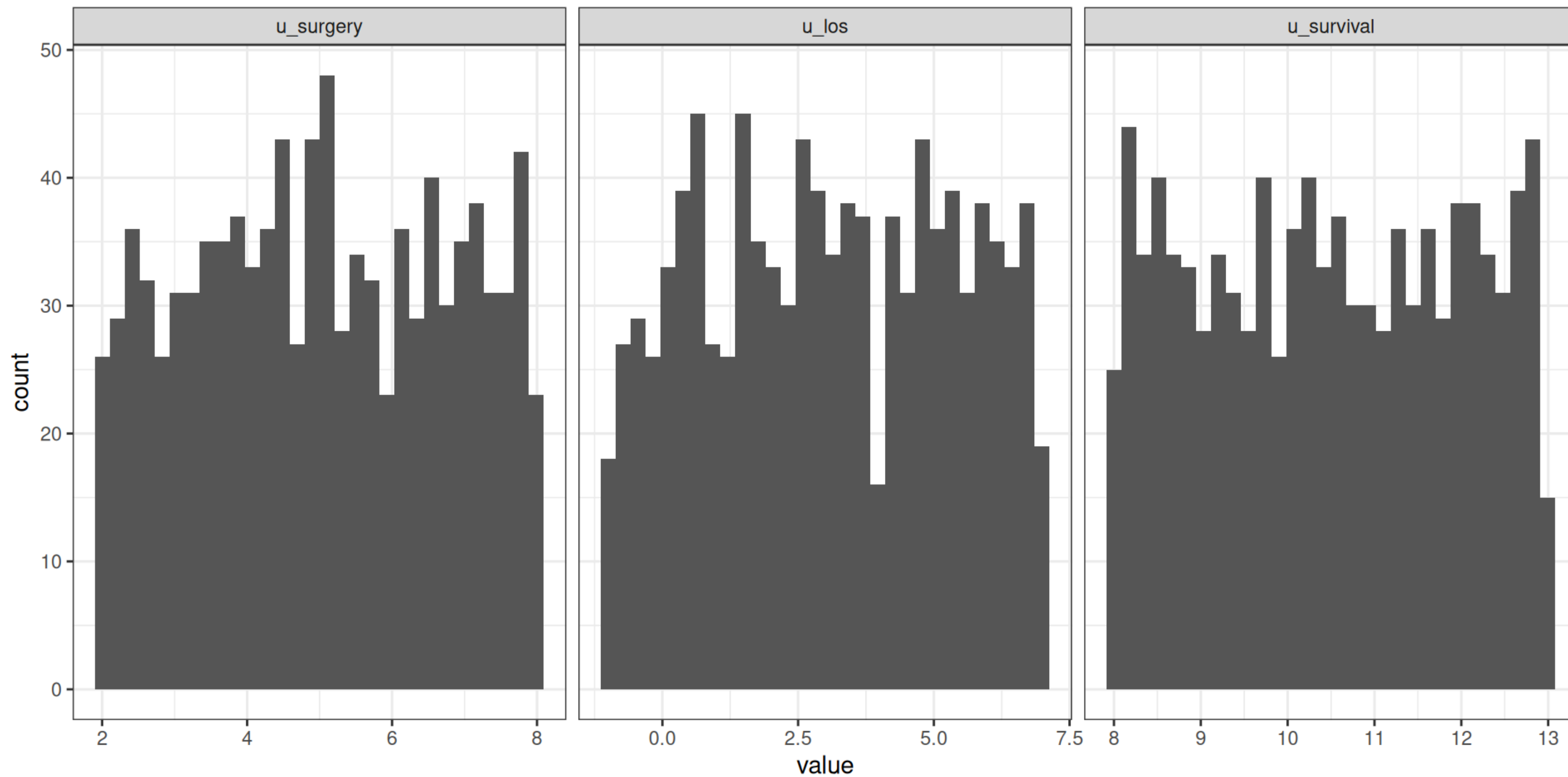


Figure 1: 1000 random samples of U

A Probabilistic Causal Model is a SCM with a distribution over U

```
1 sample_pcm <- function() {  
2   U <- sample_u()  
3   V <- scm1(U[['u_surgery']], U[['u_loss']], U[['u_survival']])  
4   c(U, V)  
5 }  
6  
7 sample_pcm()
```

u_surgery	u_loss	u_survival	surgery	loss	survival
2.059217	2.897873	10.967945	2.059217	4.957091	11.806601

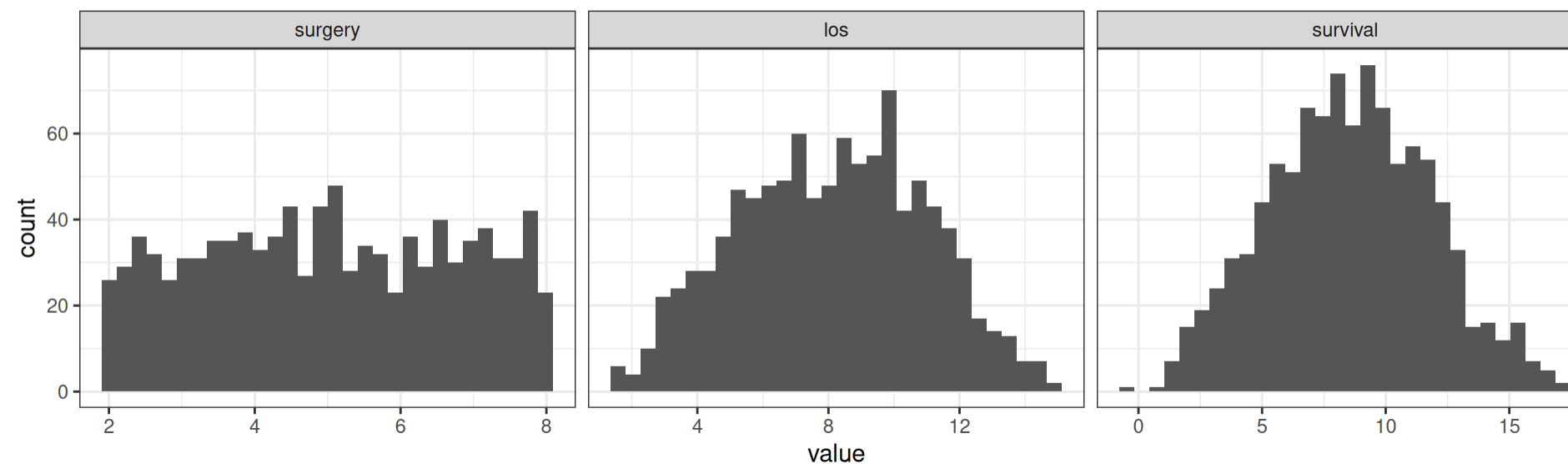


Figure 2: Realisations of endogenous variables V over random samples of U in [Figure 1](#)

Calculating a treatment effect in a fully specified probabilistic causal model

- take random samples from U , push forward through `submodel7` and `submodel3`

```
1 # N = 1e3
2 # us <- map(1:N, ~sample_u())
3
4 v3s <- map(us, ~do.call(submodel3, as.list(.x)))
5 v7s <- map(us, ~do.call(submodel7, as.list(.x)))
6
7 v3df <- v3s |> map(~data.table(t(.x))) |> rbindlist()
8 v7df <- v7s |> map(~data.table(t(.x))) |> rbindlist()
9 v3df[, idx:=.I]
10 v7df[, idx:=.I]
11
12 dfa <- rbindlist(list(
13   scm1=vdf,
14   submodel3=v3df,
15   submodel7=v7df
16 ), idcol='model')
17
18 dfa[, list(mean_survival=mean(survival)). bv="model"]
```

	model	mean_survival
	<char>	<num>
1:	scm1	8.544598
2:	submodel3	3.437454
3:	submodel7	7.437454

Identification

Recap of definitions

- *Structural Causal model*:
 - endogenous variables V
 - exogenous (noise) variables U
 - deterministic functions $f_i(pa_i, u_i)$
- *Effect of Action* $do(T = t)$: *submodel* where f_T replaced with fixed value t
- *Probabilistic Causal Model*: SCM + distribution over U

In the real world

- knowing the SCM is a super-power: you basically know everything relevant about the system, but in the real world:
 - we do not observe U
 - we typically do not know $f_{_}$
 - we may be willing to place *assumptions* on $f_{_}$ (e.g. generalized linear models)
 - we are presented with realizations V_i of this SCM over a random sample of U
- we may be interested in knowing:
 1. what is the expected survival time if we *always* admit patients for exactly 7 days?

When and how might we learn the answer to such questions?

Identification

Causal effect identification:

Definition 3.2.3 (Identifiability)

Let $Q(M)$ be any computable quantity of a model M .

We say that Q is **identifiable** in a class $\mathbb{M}$ of models if, for any pairs of models M_1 and M_2 from $\mathbb{M}$,

$Q(M_1) = Q(M_2)$ whenever $P_{M_1}(y) = P_{M_2}(y)$.

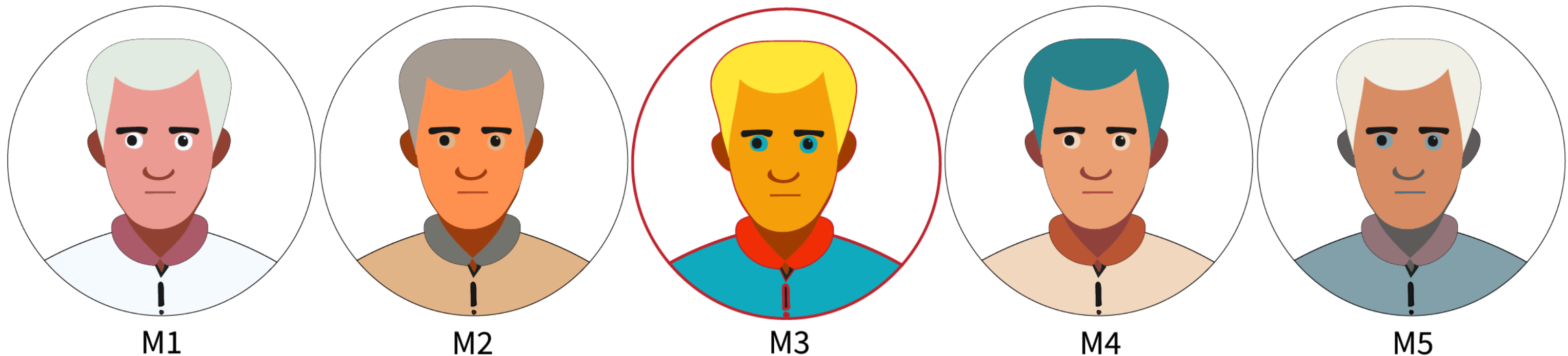
If our observations are limited and permit only a partial set F_M of features (of $P_M(y)$) to be estimated,

we define Q to be identifiable from F_M if $Q(M_1) = Q(M_2)$ whenever $F_{M_1} = F_{M_2}$.

Identification in pictures

Someone killed the priest (†), we want to know who-dunnit (= Q)

Based on prior knowledge we have 5 *suspects* (all the SCMs compatible with our DAG)

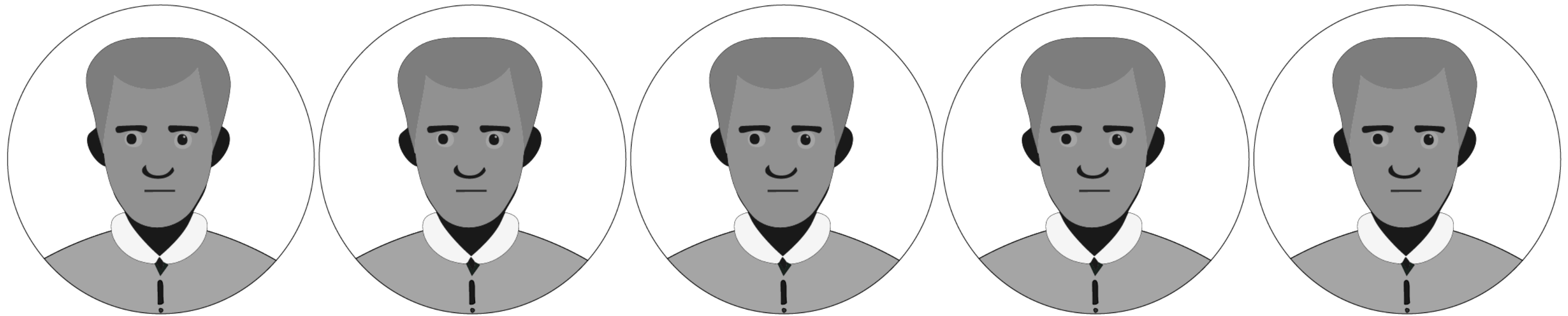


If we had full data, we would know it was M_3

Identification in pictures

Someone killed the priest (†) , we want to know who-dunnit (= Q)

Based on prior knowledge on 5 suspects (all the SCMs compatible with our DAG)



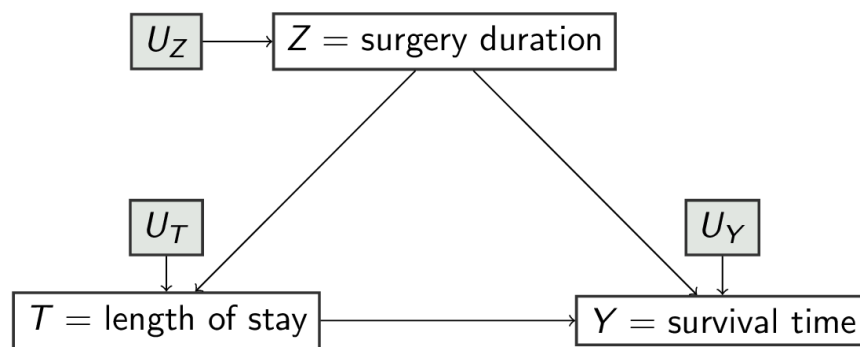
If we had full data, we would have know it was M_3

Unfortunately, it was dark an we only got a gray-scale image of the perpetrator

All our suspects (models) have the same partial observations but different answers to our question

Based on *observed data* and *assumptions* we cannot *identify* the answer to our question Q ,
i.e. multiple models with different answers for Q fit the observed data equally well

Not identified in a concrete example



The backdoor adjustment in this DAG means the correct estimand is:

$$P(Y|\text{do}(T)) = \sum_Z P(Y|T, z)P(Z = z)$$

- If we did not observe Z , we could still come up with a latent-variable model for Z and a model for $Y|T, Z$ and get a value.
- However, we can formulate multiple distinct latent variable models that each yield a different treatment effect (i.e. the output of the estimand)
- But these latent variable models all fit the *observed* data equally well
- So we cannot identify the treatment effect

Numeric example: same observations, two SCMs

Binary simplification of the slide's DAG: $Z \rightarrow T, Z \rightarrow Y, T \rightarrow Y$.

Assume Z is latent. Both models have

$$P(Z = 1) = 0.5, \quad P(T = 1|Z = 0) = 0.2, \quad P(T = 1|Z = 1) = 0.8$$

Model M_A

$P(Y = 1 T, Z)$	$Z = 0$	$Z = 1$
$T = 0$	0.20	1.00
$T = 1$	0.00	0.80

Model M_B

$P(Y = 1 T, Z)$	$Z = 0$	$Z = 1$
$T = 0$	0.40	0.20
$T = 1$	0.80	0.60

Both imply the same observed data over (T, Y) :

$$P(T = 1) = 0.5, \quad P(Y = 1|T = 1) = 0.64, \quad P(Y = 1|T = 0) = 0.36$$

Same observations, different causal answers

If Z were observed, the backdoor estimand would be

$$P(Y = 1 | \text{do}(T = t)) = \sum_z P(Y = 1 | T = t, Z = z) P(Z = z)$$

For M_A :

$$P(Y = 1 | \text{do}(T = 1)) = 0.40, \quad P(Y = 1 | \text{do}(T = 0)) = 0.60$$

$$\text{ATE}_{M_A} = -0.20$$

For M_B :

$$P(Y = 1 | \text{do}(T = 1)) = 0.70, \quad P(Y = 1 | \text{do}(T = 0)) = 0.30$$

$$\text{ATE}_{M_B} = 0.40$$

Model A vs Model B

Same observed $P(T, Y)$, different $Q(M)$.

That is **not identified**.

Seeing is not doing

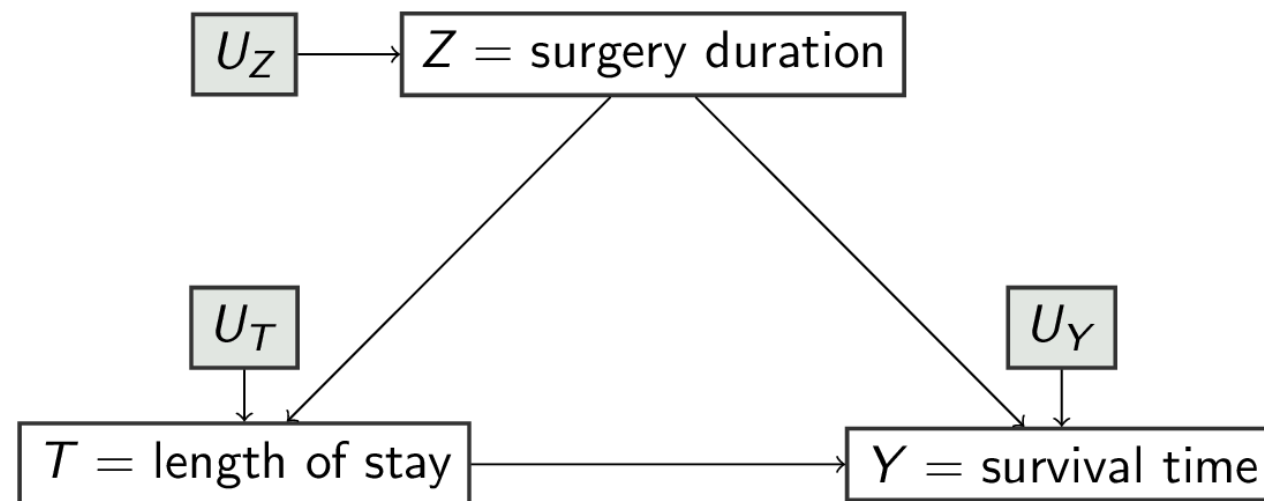
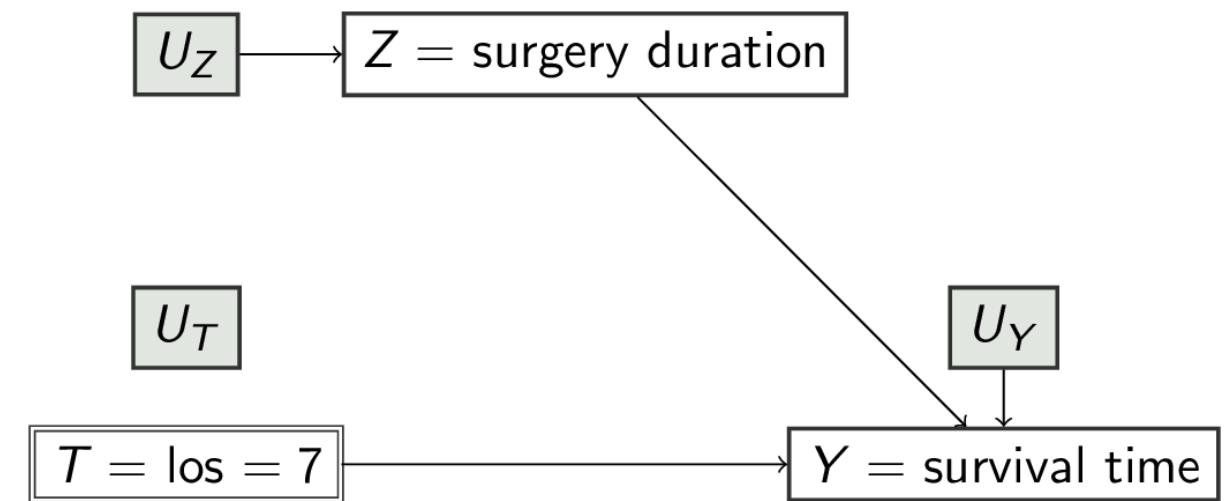


Figure 3:

$$P(Y|T) = \sum_z P(Y|T, z)P(Z = z|T)$$



$$\begin{aligned} P(Y|T) &= \sum_z P(Y|T, z)P(Z = z|T) \\ &=^2 \sum_z P(Y|T, z)P(Z = z) \end{aligned}$$

Figure 4: ² because in the intervened DAG, Z is independent of T

- $P(Y|\text{do}(T)) \neq P(Y|T)$ is Pearl's definition of confounding (def 6.2.1)

Another path to identification: parametric assumptions

- for example:
 - assumption 1: $\mathbb{M}_1$, all SCMs with same DAG
 - assumption 2: $\mathbb{M}_2$ SCMs with linear functions and Gaussian error terms
 - assumption 1+2: $\mathbb{M} = \mathbb{M}_1 \cap \mathbb{M}_2$ (DAG + linear gaussian)
- many more effects are identified in this setting
 - settings with unobserved confounding and positivity violations may still be identified
- caveats:
 - much harder to determine identifyability (no analogue of backdoor-rule)
 - prefer weaker assumptions over stronger assumption

Defining counterfactuals and the causal hierarchy (of questions)

Counterfactuals

- all of the above can be achieved with DAGs, but we haven't used SCMs *unique power* yet: counterfactuals
- RCT / DAG questions: *What is the expected survival if we keep all patients in the hospital for 7 days?*

Take it one level higher: counterfactuals

For patient Adam we had this data:

- surgery duration: 4 hours
- length of stay: 3 days
- survival: 4 years

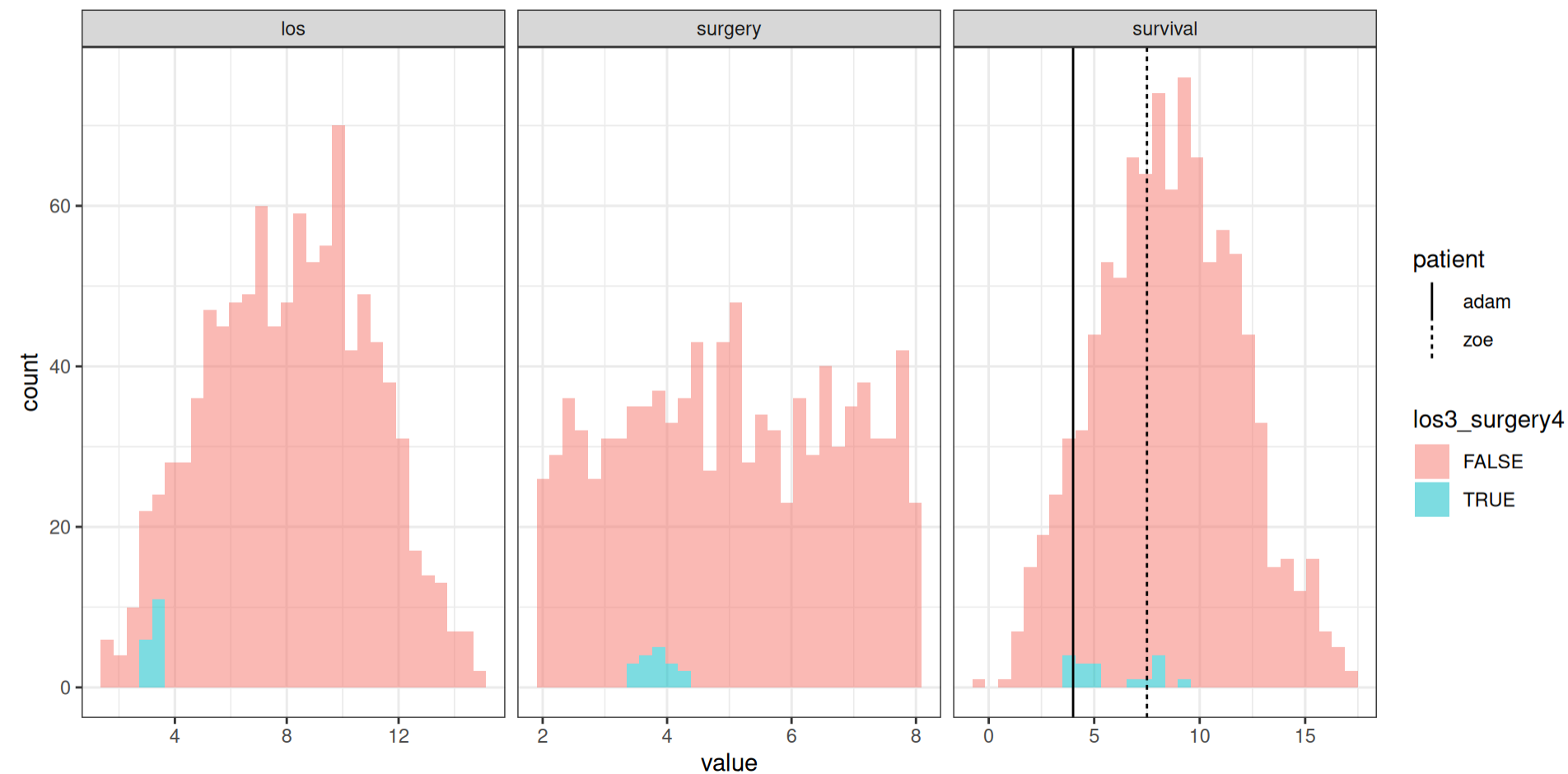
For patient Zoe we had this data:

- surgery duration: 4 hours
- length of stay: 3 days
- survival: 7.5 years

- we do not observe Adam's/Zoe's **U**
- What would the expected survival have been had Adam/Zoe been kept in the hospital for 7 days?

Adam versus Zoe

- Average causal effects in subgroup with `surgery=4`:
 - 3-days LOS: 5.2
 - 7-days LOS: 9.2



- what do we expect for Adam and Zoe if they would have been kept in the hospital for 7 days?

Computing counterfactuals with SCMs

- Given our information on the structural equation for `survival` (Section 2.2):

$$\text{survival} = \text{los} - 2 * \text{surgery} + u_{\text{survival}}$$

- and observed values on Adam's and Zoe's `surgery` AND `survival` following `los=3`
- we can compute their individual u_{survival} :

patient	surgery	los	survival	u_survival	survival7
Adam	4	3	4	9	8
Zoe	4	3	7.5	12.5	11.5

- and (counterfactual) survival under 7 days LOS

Computing counterfactuals

- notation: $P(Y_{t'} = y' | T = t, Y = y)$ where $Y_{t'}$ means “set $T = t'$ through intervention”
- this is called a *counterfactual* because it's a probability statement incorporating two incompatible worlds: the world where $T = t$ and the world where $T = t'$
- steps:
 1. Abduction (update $P(U)$ from observed evidence)
 2. Action (modify the treatment)
 3. Prediction (calculate outcomes in submodel, putting in the updated $P(U)$)

Pearl's Causal Hierarchy (of questions)

If you have data to solve the upper, you can solve the lower ranks too ([Bareinboim et al. 2022](#))

1. counterfactuals
2. interventions
3. associations

Where do we get this knowledge from?

- not from observational data
- not from RCTs
- from assumptions
- can get bounds from combinations of RCT data and observational data
- caveat: some say the hierarchy is upside down because you go further away from data and closer to unverifiable assumptions the ‘higher’ you get

Not covered but also possible:

- DAGs:
 - *soft intervention*: don't set treatment to fixed value but replace function with other function of variables
 - *selection diagrams / transportability*: ask whether effects learned in one population apply to another
 - express patterns for missing data by including missingness indicators
- SCMs:
 - probability of sufficiency
 - probability of necessity
 - individual-level counterfactual explanations
 - counterfactual fairness

References

Bareinboim, Elias, Juan Correa, Duligur Ibeling, and Thomas Icard. 2022. “On Pearl’s Hierarchy and the Foundations of Causal Inference (1st Edition).” In *Probabilistic and Causal Inference: The Works of Judea Pearl*, edited by Hector Geffner, Rita Dechter, and Joseph Halpern. ACM Books.

Pearl, Judea, ed. 2009. “The Logic of Structure-Based Counterfactuals.” In *Causality*, 2nd ed. Cambridge University Press.
<https://doi.org/10.1017/CBO9780511803161.009>.

